Unit II: Set Theory (Part A)

Sets, Subsets, and Set Operations

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Contents

PART 1: Sets — The Basics 4

1 What is a Set? 4
- 1.1 Why Do We Need Sets? . . . 4
- 1.2 What Does "Well-Defined" Mean? . . . 4
- 1.3 Notation . . . 5
- 1.4 Two Ways to Write a Set . . . 6
- 1.5 Cardinality . . . 6

2 Types of Sets 7

PART 2: Subsets and Power Set 9

3 Subsets 9
- 3.1 Step-by-Step Check . . . 9
- 3.2 Three Golden Rules of Subsets . . . 10

4 Proper Subsets 10

5 Power Set 11

PART 3: Complement of a Set 12

6 Complement 12
- 6.1 Properties of Complement . . . 12
- 6.2 Venn Diagram for Complement . . . 13

PART 4: Union, Intersection, Difference 14

7 Union of Sets ($A \cup B$) 14
- 7.1 Step-by-Step Method . . . 14

8 Intersection of Sets ($A \cap B$) 15
- 8.1 Disjoint Sets . . . 16

9 Difference of Sets ($A - B$) 16

PART 5: De Morgan's Laws 18

10 De Morgan's Laws 18
- 10.1 Intuitive Meaning . . . 18
- 10.2 Verification with Numbers . . . 18
- 10.3 Venn Diagram for De Morgan's Law 1 . . . 20

11 Counting with Union (2 Sets) **20**

12 Counting with Complement **21**

13 Counting with Difference **21**

14 Counting Cartesian Products **21**

15 Venn Diagram Word Problems **21**

16 In-Class Activities: Counting **22**

17 In-Class Activities **23**

18 Summary **25**

19 Practice Problems **25**

PART 1: Sets — The Basics

1 What is a Set?

Definition: A **set** is a *well-defined* collection of *distinct* objects. The objects in a set are called its **elements** or **members**.

1.1 Why Do We Need Sets?

Before we see formal examples, think about everyday life:

- A **cricket team** is a group of 11 players.
- A **fruit basket** contains specific fruits: apple, banana, mango.
- The **natural numbers** $\{1, 2, 3, \ldots\}$ form a collection.
- The **set of vowels** in English: a, e, i, o, u.

In each case, we have a collection of distinct objects grouped together by a clear rule. That is exactly what a **set** is.

1.2 What Does "Well-Defined" Mean?

A collection is well-defined if, given any object, we can **unambiguously decide** whether that object belongs to the collection or not.

Example 1 (Well-defined sets):

(a) The set of all **players in the Indian cricket team** (2026 squad).
Well-defined? Yes. Any player is either in the squad or not.

(b) The set of all **fruits containing Vitamin C**.
Well-defined? Yes. A fruit either has Vitamin C or it doesn't.

(c) The set of all **prime numbers less than 20**.
Well-defined? Yes. We can list them: $2, 3, 5, 7, 11, 13, 17, 19$.

(d) The set of all **sports played in the Olympics**.
Well-defined? Yes. The IOC publishes an official list.

Example 2 (NOT well-defined — these are NOT sets):

(a) The collection of all **tasty dishes**.
Why not? "Tasty" is subjective — it depends on personal preference.

(b) The collection of all **great cricket players**.
Why not? "Great" is not objectively defined.

(c) The collection of all **big numbers**.

Why not? Is 1000 big? Is 1 lakh big? Is 1 crore big? No clear cutoff.

1.3 Notation

Symbol	Meaning	Example
A, B, S, T	Sets are named with capital letters	$A = \{1, 2, 3\}$
$\{\ \}$	Curly braces enclose the elements	{apple, banana}
$\in$	"is an element of"	mango $\in$ Fruits
$\notin$	"is NOT an element of"	carrot $\notin$ Fruits
$n(A)$ or $\lvert A\rvert$	Cardinality (number of elements)	$n(\{1, 2, 3\}) = 3$
$\varnothing$	Empty set (no elements)	$\{\} = \varnothing$
U	Universal set (all possible elements)	See below

Example 3 (Sports — Using $\in$ and $\notin$):
Let T be a set of the Indian cricket team players:

$$T = \{\text{Rohit, Virat, Mandhana, Harmanpreet, Gill, Bumrah}\}$$

Then:

$$\text{Virat} \in T, \quad \text{Mandhana} \in T, \quad \text{Bumrah} \in T$$

$$\text{Messi} \notin T \quad \text{(football player, not in cricket team)}$$

$$\text{Dhoni} \notin T$$

Example 4 (Food — Using $\in$ and $\notin$):
Let the set of fruits in a basket be:

$$F = \{\text{apple, banana, mango, orange, grapes}\}$$

Then:

$$\text{apple} \in F, \quad \text{mango} \in F$$

$$\text{potato} \notin F, \quad \text{tomato} \notin F$$

Example 5 (Number Systems — Using $\in$ and $\notin$):
Let $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$ be the set of natural numbers.

- $5 \in \mathbb{N}$ ✓
- $100 \in \mathbb{N}$ ✓
- $-3 \notin \mathbb{N}$ × (negative numbers not natural)
- $0 \notin \mathbb{N}$ (in this course, $\mathbb{N}$ starts at 1)

- $\frac{1}{2} \notin \mathbb{N} \times$ (fractions are not natural)

1.4 Two Ways to Write a Set

(A) Roster Method (Listing): List all elements inside braces.

$$A = \{2, 4, 6, 8, 10\}$$

$$\text{Sports} = \{\text{cricket, football, hockey, tennis, badminton}\}$$

(B) Set-Builder Method (Rule): Describe the rule that defines membership.

$$A = \{x \mid x \text{ is an even number less than } 12\}$$

$$B = \{x \mid x \text{ is a fruit with Vitamin C}\}$$

Example 6 (Converting Between Notations):
Roster → Set-Builder:

- $\{2, 3, 5, 7, 11\} = \{x \mid x \text{ is a prime number less than } 12\}$
- $\{\text{apple, banana, mango}\} = \{x \mid x \text{ is a tropical fruit in my basket}\}$
- $\{\text{chess, carrom, ludo}\} = \{x \mid x \text{ is an indoor board game}\}$

Set-Builder → Roster:

- $\{x \mid x \text{ is a day starting with T}\} = \{\text{Tuesday, Thursday}\}$
- $\{x \mid x \in \mathbb{N},\ x < 8,\ x \text{ is odd}\} = \{1, 3, 5, 7\}$
- $\{x \mid x \text{ is a vowel in English}\} = \{a, e, i, o, u\}$

1.5 Cardinality

Cardinality of a set A, written $n(A)$ or $|A|$, is the number of distinct elements in A.

Example 7 (Cardinality — Sports & Food & Numbers):

$$n(\{\text{Rohit, Virat, Gill, Bumrah}\}) = 4$$

$$n(\{\text{apple, banana, mango}\}) = 3$$

$$n(\{2, 4, 6, 8, 10\}) = 5$$

$$n(\{\}) = 0 \quad \text{(empty set)}$$

2 Types of Sets

Type	Definition	Example
Empty (Null) Set $\varnothing$	Contains no elements	Students who scored 200/100
Singleton Set	Contains exactly one element	{the current Indian captain}
Finite Set	Countable number of elements	$\{1, 2, 3, 4, 5\}$
Infinite Set	Unlimited elements	$\mathbb{N} = \{1, 2, 3, \ldots\}$
Equal Sets	Same elements in any order	$\{1, 2, 3\} = \{3, 2, 1\}$
Universal Set U	All possible elements under study	All sports played worldwide

Example 8 (Empty Set — Food):
Let $A = \{x \mid x \text{ is a fruit that tastes both sweet and bitter at the same time}\}$. This set is empty because no fruit is both sweet and bitter in the same bite. So $A = \varnothing$.

Another example: $B = \{x \mid x \text{ is a player who has won 10 World Cups}\}$. No player has won 10 World Cups, so $B = \varnothing$.

Example 9 (Singleton Set — Sports):
Let $A = \{x \mid x \text{ is the current captain of the Indian men's ODI cricket team}\}$.

$$A = \{\text{Shubhman Gill}\}$$

$$n(A) = 1$$

This is a **singleton set**.

Example 10 (Finite vs Infinite — Numbers):

- $\{1, 2, 3, 4, 5\}$ is **finite** $(n = 5)$.
- $\mathbb{N} = \{1, 2, 3, 4, \ldots\}$ is **infinite**.
- $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ (integers) is **infinite**.
- $\mathbb{Q} = \left\{\frac{p}{q} \mid p, q \in \mathbb{Z},\ q \neq 0\right\}$ (rationals) is **infinite**.

Example 11 (Equal vs Equivalent):

$$A = \{\text{cricket, football, hockey}\}$$

$$B = \{\text{hockey, cricket, football}\}$$

$$C = \{10, 20, 30\}$$

- A and B are **equal** (same elements, order irrelevant).
- A and C are **equivalent** ($n(A) = n(C) = 3$) but not equal.

Example 12 (Universal Set — Cricket Tournament):
In an ICC Cricket World Cup:

$$U = \{\text{all 10 participating teams}\}$$

$$A = \{\text{Asian teams}\} = \{\text{India, Pakistan, Sri Lanka, Bangladesh, Afghanistan}\}$$

$B = \{\text{Test-playing nations with 5+ World Cup wins}\} = \{\text{Australia, India, West Indies}\}$

Here $A \subseteq U$ and $B \subseteq U$.

PART 2: Subsets and Power Set

3 Subsets

Definition: Set A is a **subset** of set B if *every element* of A is also an element of B. We write:

$$A \subseteq B$$

Read as: "A is a subset of B" or "A is contained in B."

3.1 Step-by-Step Check

Step 1. List all elements of A.

Step 2. For each element, check: is it in B?

Step 3. If **every** element of A is in B, then $A \subseteq B$.

Step 4. If **even one** element is missing, then $A \not\subseteq B$.

Example 13 (Sports — Subset Check):
Let:

$$U = \{\text{cricket, football, hockey, tennis, badminton, chess}\}$$

$$A = \{\text{cricket, football, hockey}\}$$

$$B = \{\text{tennis, badminton}\}$$

Is $A \subseteq U$? Yes — all elements of A are in U. ✓
Is $B \subseteq U$? Yes. ✓
Is $A \subseteq B$? No — cricket is in A but not in B. ×
Is $B \subseteq A$? No — tennis is in B but not in A. ×

Example 14 (Food — Subset Check):
Let:

$$U = \{\text{apple, banana, mango, potato, tomato, carrot, spinach}\}$$

$$F = \{\text{apple, banana, mango}\} \quad \text{(fruits)}$$

$$V = \{\text{potato, tomato, carrot, spinach}\} \quad \text{(vegetables)}$$

Is $F \subseteq U$? Yes. ✓
Is $V \subseteq U$? Yes. ✓
Is $F \subseteq V$? No — apple is not a vegetable. ×

Example 15 (Number Systems — The Classic Chain):

$$\mathbb{N} = \{1, 2, 3, \ldots\} \quad \text{(Natural numbers)}$$

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\} \quad \text{(Integers)}$$

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z},\ q \neq 0 \right\} \quad \text{(Rationals)}$$

$$\mathbb{R} = \text{all real numbers}$$

Then:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

This is one of the most beautiful chains in mathematics — each number system *contains* the previous one.

3.2 Three Golden Rules of Subsets

1. **Every set is a subset of itself:** $A \subseteq A$ is always true.
2. **The empty set is a subset of every set:** $\varnothing \subseteq A$ is always true.
3. **Mutual inclusion means equality:** If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

4 Proper Subsets

Definition: A is a **proper subset** of B if $A \subseteq B$ AND $A \neq B$. We write:

$$A \subset B$$

This means A is a *strictly smaller* set that fits inside B.

Example 16 (Proper Subset — Numbers):

$$\{1, 2\} \subset \{1, 2, 3\} \quad \text{(proper subset)}\checkmark$$

$$\{1, 2, 3\} \subset \{1, 2, 3\} \quad \text{(NOT proper — they are equal)}$$

$$\mathbb{N} \subset \mathbb{Z} \quad \text{(proper subset)}\checkmark$$

$$\varnothing \subset \{1\} \quad \text{(proper subset)}\checkmark$$

Example 17 (Proper Subset — Food):

Let:

$$\text{All Dishes} = \{\text{pizza, biryani, dosa, paneer, salad}\}$$

$$\text{Vegetarian Dishes} = \{\text{dosa, paneer, salad}\}$$

Then:

$$\text{Vegetarian Dishes} \subset \text{All Dishes} \quad \checkmark$$

But:

$$\text{All Dishes} \not\subset \text{Vegetarian Dishes}$$

because pizza and biryani are not vegetarian.

5 Power Set

Definition: The **power set** of A, written $\mathcal{P}(A)$, is the set of **all possible subsets** of A, including $\varnothing$ and A itself.

Formula:

$$\boxed{n(\mathcal{P}(A)) = 2^{n(A)}}$$

Example 18 (Power Set — Food):

Let $A = \{\text{apple, banana}\}$. Then $n(A) = 2$, so $n(\mathcal{P}(A)) = 2^2 = 4$.

$$\mathcal{P}(A) = \{\varnothing,\ \{\text{apple}\},\ \{\text{banana}\},\ \{\text{apple, banana}\}\}$$

Interpretation: These are all possible ways to select fruits from a basket containing 2 fruits:

- Take nothing ($\varnothing$)
- Take only apple
- Take only banana
- Take both

Example 19 (Power Set — Sports):

Let $S = \{\text{chess, carrom, ludo}\}$. Then $n(S) = 3$, so $n(\mathcal{P}(S)) = 2^3 = 8$.

$$\mathcal{P}(S) = \left\{ \begin{array}{c} \varnothing,\ \{\text{chess}\},\ \{\text{carrom}\},\ \{\text{ludo}\}, \\ \{\text{chess, carrom}\},\ \{\text{chess, ludo}\},\ \{\text{carrom, ludo}\}, \\ \{\text{chess, carrom, ludo}\} \end{array} \right\}$$

Interpretation: These are all possible teams you can form from 3 board games (including an empty team).

Example 20 (Power Set — Numbers):

Let $A = \{2, 4, 6\}$. Then $n(A) = 3$, so $n(\mathcal{P}(A)) = 8$.

The subsets are:

$$\varnothing,\ \{2\},\ \{4\},\ \{6\},\ \{2, 4\},\ \{2, 6\},\ \{4, 6\},\ \{2, 4, 6\}$$

Check: 1 empty + 3 singletons + 3 pairs + 1 full set = 8. ✓

PART 3: Complement of a Set

6 Complement

Definition: The **complement** of A (written A' or A^c) is the set of all elements in the universal set U that are **NOT** in A:

$$\boxed{A' = \{x \in U \mid x \notin A\}}$$

Equivalently, $A' = U - A$.

Example 21 (Complement — Numbers):
Let:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\} \quad \text{(even numbers)}$$

Then:

$$A' = \{1, 3, 5, 7, 9\} \quad \text{(odd numbers)}$$

Key insight: The complement of even numbers is odd numbers. ✓

Example 22 (Complement — Food):
Let the universal set of dishes in a restaurant be:

$$U = \{\text{pizza, biryani, dosa, salad, paneer, burger, pasta}\}$$

$$\text{Veg} = \{\text{dosa, salad, paneer, pasta}\}$$

Then the complement of Veg is:

$$\text{Veg}' = \{\text{pizza, biryani, burger}\} \quad \text{(non-vegetarian dishes)}$$

Example 23 (Complement — Sports):
Let:

$$U = \{\text{cricket, football, hockey, tennis, badminton, chess, carrom}\}$$

$$\text{Outdoor} = \{\text{cricket, football, hockey, tennis}\}$$

Then:

$$\text{Outdoor}' = \{\text{badminton, chess, carrom}\} \quad \text{(indoor games)}$$

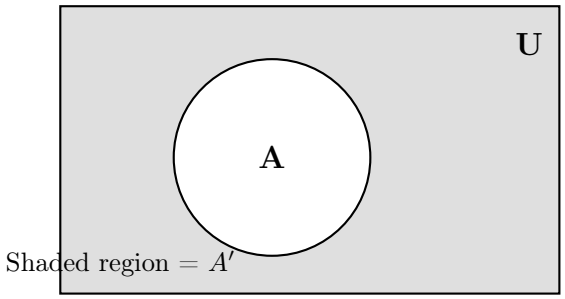
6.1 Properties of Complement

1. **Double Complement:** $(A')' = A$
2. **Complement of U:** $U' = \varnothing$
3. **Complement of $\varnothing$:** $\varnothing' = U$
4. **Union with Complement:** $A \cup A' = U$
5. **Intersection with Complement:** $A \cap A' = \varnothing$

Example 24 (Verifying Properties):
Using the number example above ($U = \{1, \ldots, 10\}$, $A = \{2, 4, 6, 8, 10\}$):

- $A' = \{1, 3, 5, 7, 9\}$
- $(A')' = \{2, 4, 6, 8, 10\} = A$ ✓
- $A \cup A' = \{1, 2, 3, \ldots, 10\} = U$ ✓
- $A \cap A' = \varnothing$ ✓

6.2 Venn Diagram for Complement



The **shaded region** (everything outside A but inside U) is the complement A'.

PART 4: Union, Intersection, Difference

7 Union of Sets ($A \cup B$)

Definition: The **union** of A and B, written $A \cup B$, is the set of *all* elements that are **either** in A, **or** in B:

$$\boxed{A \cup B = \{x \mid x \in A \text{ or } x \in B\}}$$

The word "or" is **inclusive** — it means x can be in A, in B, or in both.

7.1 Step-by-Step Method

Step 1. Write down all elements of A.

Step 2. Write down all elements of B.

Step 3. Combine into one list.

Step 4. Remove duplicates.

Example 25 (Union — Food):
Let:

$$A = \{\text{apple, banana, mango, grapes}\}$$

$$B = \{\text{grapes, orange, papaya, banana}\}$$

Then:

$$A \cup B = \{\text{apple, banana, mango, grapes, orange, papaya}\}$$

$$n(A \cup B) = 6$$

Note: grapes and banana appear in both, so they are listed only once.

Example 26 (Union — Sports Teams):
Two sports clubs have the following players:

$$\text{Club X} = \{\text{Aarav, Bhavna, Chirag, Divya}\}$$

$$\text{Club Y} = \{\text{Chirag, Divya, Esha, Farhan}\}$$

The union is:

$$\text{Club X} \cup \text{Club Y} = \{\text{Aarav, Bhavna, Chirag, Divya, Esha, Farhan}\}$$

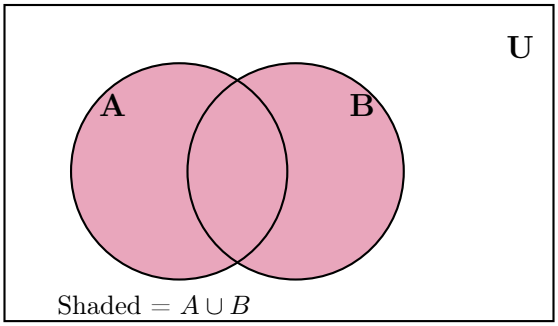
Interpretation: This is the set of *all players* who play for **at least one of the two clubs**.

Example 27 (Union — Number Systems):

$$A = \{1, 2, 3, 4\} \quad \text{(natural numbers less than 5)}$$

$$B = \{3, 4, 5, 6\} \quad \text{(even/odd mix)}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$



8 Intersection of Sets ($A \cap B$)

Definition: The **intersection** of A and B, written $A \cap B$, is the set of elements that are in **both** A **and** B:

$$\boxed{A \cap B = \{x \mid x \in A \text{ and } x \in B\}}$$

Example 28 (Intersection — Food):
Using $A = \{\text{apple, banana, mango, grapes}\}$ and $B = \{\text{grapes, orange, papaya, banana}\}$:

$$A \cap B = \{\text{banana, grapes}\}$$

Interpretation: Banana and grapes are the fruits common to both baskets.

Example 29 (Intersection — Sports):
Using the club example:

$$\text{Club X} = \{\text{Aarav, Bhavna, Chirag, Divya}\}$$

$$\text{Club Y} = \{\text{Chirag, Divya, Esha, Farhan}\}$$

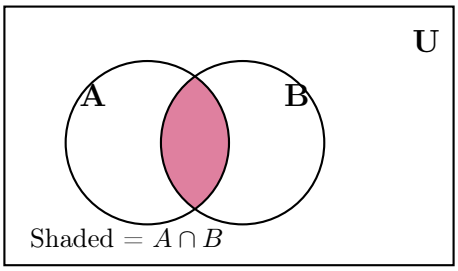
$$\text{Club X} \cap \text{Club Y} = \{\text{Chirag, Divya}\}$$

Interpretation: Chirag and Divya play for **both** clubs.

Example 30 (Intersection — Number Systems):

$$A = \{2, 4, 6, 8, 10\} \quad \text{(even numbers up to 10)}$$
$$B = \{3, 6, 9\} \quad \text{(multiples of 3 up to 10)}$$
$$A \cap B = \{6\} \quad \text{(numbers that are both even AND multiples of 3)}$$

The only such number up to 10 is 6. ✓



8.1 Disjoint Sets

Two sets are **disjoint** if they have no elements in common:

$$A \cap B = \varnothing$$

Example 31 (Disjoint — Food):

$$\{\text{apple, banana, mango}\} \cap \{\text{potato, onion, garlic}\} = \varnothing$$

Fruits and vegetables in this example are **disjoint sets**.

9 Difference of Sets ($A - B$)

Definition: The **difference** $A - B$ (also written $A \setminus B$) is the set of elements that are in A but **not** in B:

$$\boxed{A - B = \{x \mid x \in A \text{ and } x \notin B\}}$$

Important: $A - B \neq B - A$ in general.

Note: To obtain the set $A - B$; we need to remove all elements of B from the set A. Remaining part in A is $A - B$.

Example 32 (Difference — Food):

Using $A = \{\text{apple, banana, mango, grapes}\}$ and $B = \{\text{grapes, orange, papaya, banana}\}$:

$$A - B = \{\text{apple, mango}\} \quad \text{(in A but not B)}$$
$$B - A = \{\text{orange, papaya}\} \quad \text{(in B but not A)}$$

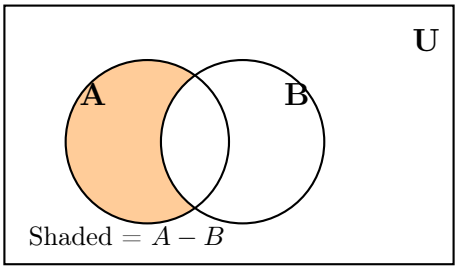
Notice: $A - B \neq B - A$. ✓

Example 33 (Difference — Sports):

$$\text{Club X} - \text{Club Y} = \{\text{Aarav, Bhavna}\} \quad \text{(only in Club X)}$$
$$\text{Club Y} - \text{Club X} = \{\text{Esha, Farhan}\} \quad \text{(only in Club Y)}$$

Example 34 (Difference — Numbers):

$$A = \{1, 2, 3, 4, 5\}$$
$$B = \{4, 5, 6, 7\}$$
$$A - B = \{1, 2, 3\}$$
$$B - A = \{6, 7\}$$



PART 5: De Morgan's Laws

10 De Morgan's Laws

De Morgan's Laws: For any two sets A and B with universal set U:

Law 1:

$$\boxed{(A \cup B)' = A' \cap B'}$$

"The complement of the union is the intersection of the complements."

Law 2:

$$\boxed{(A \cap B)' = A' \cup B'}$$

"The complement of the intersection is the union of the complements."

10.1 Intuitive Meaning

- **Law 1:** "Not (A or B)" = "(Not A) and (Not B)"
- **Law 2:** "Not (A and B)" = "(Not A) or (Not B)"

10.2 Verification with Numbers

Example 35 (Verify Law 1 with Numbers):

Let:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$
$$A = \{1, 2, 3, 4\}$$
$$B = \{3, 4, 5, 6\}$$

Left-hand side: $(A \cup B)'$

1. $A \cup B = \{1, 2, 3, 4, 5, 6\}$
2. $(A \cup B)' = \{7, 8\}$

Right-hand side: $A' \cap B'$

1. $A' = \{5, 6, 7, 8\}$
2. $B' = \{1, 2, 7, 8\}$
3. $A' \cap B' = \{7, 8\}$

Conclusion: $(A \cup B)' = A' \cap B'$ ✓

Example 36 (Verify Law 2 with Numbers):

Using the same sets:

Left-hand side: $(A \cap B)'$

1. $A \cap B = \{3, 4\}$

2. $(A \cap B)' = \{1, 2, 5, 6, 7, 8\}$

Right-hand side: $A' \cup B'$

1. $A' = \{5, 6, 7, 8\}$
2. $B' = \{1, 2, 7, 8\}$
3. $A' \cup B' = \{1, 2, 5, 6, 7, 8\}$

Conclusion: $(A \cap B)' = A' \cup B'$ ✓

Example 37 (De Morgan with Food):
Let the universal set of dishes be:

$$U = \{\text{pizza, biryani, dosa, paneer, salad, burger}\}$$

$$\text{Spicy} = \{\text{biryani, burger}\} \quad \text{(A)}$$

$$\text{Veg} = \{\text{dosa, paneer, salad}\} \quad \text{(B)}$$

Question: What is $(\text{Spicy} \cup \text{Veg})'$?
Direct method:

$$\text{Spicy} \cup \text{Veg} = \{\text{biryani, burger, dosa, paneer, salad}\}$$

$$(\text{Spicy} \cup \text{Veg})' = \{\text{pizza}\}$$

De Morgan method:

$$\text{Spicy}' = \{\text{pizza, dosa, paneer, salad}\}$$

$$\text{Veg}' = \{\text{pizza, biryani, burger}\}$$

$$\text{Spicy}' \cap \text{Veg}' = \{\text{pizza}\}$$

Interpretation: Pizza is the only dish that is **neither spicy nor vegetarian**. Both methods give the same answer. ✓

Example 38 (De Morgan with Sports):

$$U = \{\text{cricket, football, hockey, tennis, chess, carrom}\}$$

$$\text{Outdoor} = \{\text{cricket, football, hockey, tennis}\} \quad \text{(A)}$$

$$\text{Team Sport} = \{\text{cricket, football, hockey}\} \quad \text{(B)}$$

Question: What is $(\text{Outdoor} \cap \text{Team Sport})'$?
Direct:

$$\text{Outdoor} \cap \text{Team Sport} = \{\text{cricket, football, hockey}\}$$

$$(\text{Outdoor} \cap \text{Team Sport})' = \{\text{tennis, chess, carrom}\}$$

De Morgan:

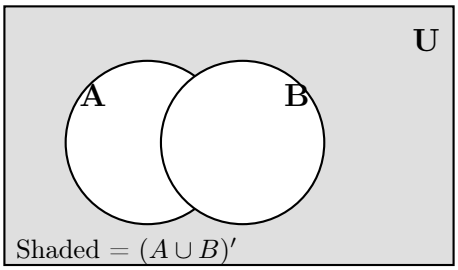
$$\text{Outdoor}' = \{\text{chess, carrom}\}$$

$$\text{Team Sport}' = \{\text{tennis, chess, carrom}\}$$

$$\text{Outdoor}' \cup \text{Team Sport}' = \{\text{tennis, chess, carrom}\}$$

Both give the same result. ✓

10.3 Venn Diagram for De Morgan's Law 1



The shaded region is **everything outside both circles** — this is the same as $A' \cap B'$.

11 Counting with Union (2 Sets)

Inclusion-Exclusion Principle (2 sets):

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

Why? When we add $n(A)$ and $n(B)$, the elements in $A \cap B$ are counted **twice**. We subtract $n(A \cap B)$ once to correct this.

Example 2 (Sports):
In a class of 40 students:

- 25 play **cricket** (A)
- 20 play **football** (B)
- 10 play **both**

How many play at least one sport?

$$n(A \cup B) = 25 + 20 - 10 = 35$$

How many play neither? Since $n(U) = 40$:

$$n(A \cup B)' = 40 - 35 = 5$$

Example 3 (Food):
In a survey of 100 people:

- 60 like **pizza** (A)
- 50 like **burger** (B)

- 30 like **both**

$$n(A \cup B) = 60 + 50 - 30 = 80$$

So 80 like **at least one**; 20 like neither.

12 Counting with Complement

$$\boxed{n(A') = n(U) - n(A)}$$

Example 5: If $n(U) = 50$ and $n(A) = 32$, then:

$$n(A') = 50 - 32 = 18$$

13 Counting with Difference

$$\boxed{n(A - B) = n(A) - n(A \cap B)}$$

Similarly, $n(B - A) = n(B) - n(A \cap B)$.

Example 6: If $n(A) = 20$ and $n(A \cap B) = 7$:

$$n(A - B) = 20 - 7 = 13$$

14 Counting Cartesian Products

$$\boxed{n(A \times B) = n(A) \times n(B)}$$

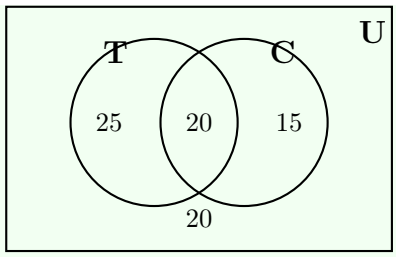
Example 7: If $n(A) = 4$, $n(B) = 5$:

$$n(A \times B) = 4 \times 5 = 20$$

.

15 Venn Diagram Word Problems

Example 8 (2-Set Venn Problem):
In a hostel of 80 students, 45 drink tea, 35 drink coffee, 20 drink both.



- Only Tea: $45 - 20 = 25$
- Only Coffee: $35 - 20 = 15$
- Both: 20
- Neither: $80 - (25 + 20 + 15) = 20$

16 In-Class Activities: Counting

Activity 1: 2-Set Counting (5 min, individual)
In a class of 50 students, 30 play cricket, 25 play badminton, 15 play both. Find:

(a) $n(C \cup B)$

(b) Number who play neither

(c) Number who play only cricket

(d) Number who play only badminton

17 In-Class Activities

Activity 1: Sport Team Subset (5 minutes, individual)

Let the universal set of sports be:

$$U = \{\text{cricket, football, hockey, tennis, badminton, chess, carrom}\}$$

Let:

$$A = \{\text{cricket, football, hockey}\} \quad \text{(outdoor team sports)}$$

$$B = \{\text{chess, carrom}\} \quad \text{(indoor board games)}$$

Determine whether each statement is **True** or **False**:

(a) $A \subseteq U$

(b) $B \subseteq U$

(c) $A \subseteq B$

(d) $B \subseteq A$

(e) $A \cap B = \varnothing$

Activity 2: Food Operations (10 minutes, pairs)

Let:

$$A = \{\text{apple, banana, mango, grapes}\}$$

$$B = \{\text{grapes, orange, banana, papaya}\}$$

Find:

(a) $A \cup B$

(b) $A \cap B$

(c) $A - B$

(d) $B - A$

(e) Is $A \cap B = \varnothing$? What does this mean?

Activity 3: Number Systems Chain (5 minutes, individual)

Arrange the following sets in increasing order of inclusion ($\subset$):

$$\mathbb{N},\ \mathbb{Z},\ \mathbb{Q},\ \mathbb{R}$$

Then give **one example** of a number:

(a) That belongs to $\mathbb{N}$

(b) That belongs to $\mathbb{Z}$ but not $\mathbb{N}$

(c) That belongs to $\mathbb{Q}$ but not $\mathbb{Z}$

(d) That belongs to $\mathbb{R}$ but not $\mathbb{Q}$

Activity 4: Power Set (10 minutes, groups of 3)

Let $A = \{\text{idli, dosa, vada}\}$ (three South Indian breakfast items).

(a) Find $n(A)$.

(b) How many subsets does A have?

(c) Write out the full power set $\mathcal{P}(A)$.

(d) How many **proper** subsets does A have?

Activity 5: De Morgan Verification (10 minutes, pairs)

Let:

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\} \quad \text{(even numbers)}$$

$$B = \{3, 6, 9\} \quad \text{(multiples of 3)}$$

Verify both De Morgan's laws:

(a) $(A \cup B)' = A' \cap B'$

(b) $(A \cap B)' = A' \cup B'$

Activity 6: True or False? (5 minutes, group discussion)

For any sets A, B, determine whether each statement is **always true**, **sometimes true**, or **never true**:

(1) $A \subseteq A$

(2) $\varnothing \subseteq A$

(3) $A \subset A$

(4) $A \cap B = B \cap A$

(5) $A - B = B - A$

(6) $A \cup A' = U$

(7) $A \cap A' = U$

(8) $(A \cup B)' = A' \cap B'$

18 Summary

Key Takeaways

1. A **set** is a well-defined collection of distinct objects.
2. Elements are written inside { } and membership is denoted by $\in$ or $\notin$.
3. Two notations: **roster** (listing) and **set-builder** (rule).
4. **Cardinality** $n(A)$ = number of elements.
5. Types: empty, singleton, finite, infinite, equal, equivalent, universal.
6. $A \subseteq B$ means every element of A is in B.
7. $A \subset B$ means $A \subseteq B$ and $A \neq B$.
8. $n(\mathcal{P}(A)) = 2^{n(A)}$.
9. A' = everything in U not in A.
10. $A \cup B$ = elements in A or B or both.
11. $A \cap B$ = elements in both A and B.
12. $A - B$ = elements in A but not B.
13. **De Morgan:** $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.
14. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
15. $n(A') = n(U) - n(A)$
16. $n(A - B) = n(A) - n(A \cap B)$
17. $n(A \times B) = n(A) \times n(B)$

19 Practice Problems

1. Define a universal set U of 6 fruits. Then create two subsets A and B. Find $n(U)$, $n(A)$, $n(B)$.

2. Convert to roster form (where possible):

   (a) $\{x \mid x \text{ is a medal colour in the Olympics}\}$

   (b) $\{x \mid x \in \mathbb{N},\ 2 \leq x \leq 8\}$

   (c) $\{x \mid x \text{ is a cricket format}\}$

3. Convert to set-builder form:

   (a) {apple, banana, mango, grapes}

   (b) $\{1, 4, 9, 16, 25\}$

   (c) {gold, silver, bronze}

4. Explain why each is NOT a set:
   (a) The collection of all tasty snacks.
   (b) The collection of all great athletes.
   (c) The collection of all big numbers.

5. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{1, 2, 3, 4, 5\}$.
   (a) Is $A \subseteq U$? Is $B \subseteq U$?
   (b) Find A' and B'.
   (c) Find $A \cup B$, $A \cap B$, $A - B$, $B - A$.
   (d) Verify De Morgan's Law 1.

6. Let $A = \{\text{apple, banana, mango}\}$.
   (a) Write $\mathcal{P}(A)$.
   (b) How many proper subsets does A have?

7. Let $A = \{\text{cricket, football}\}$, $B = \{\text{tennis, badminton}\}$.
   (a) Find $A \times B$ and $B \times A$.
   (b) Is $A \times B = B \times A$?

— End of Notes —

Unit II: Set Theory (Part B)

Cartesian Product, Relations, and Functions

MAT113 – Mathematics for Biosciences

B.Sc. Forensic Science Sem I (Batch 2026–2029)

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Contents

PART A: Cartesian Product of Sets 4

1 Ordered Pairs 4
- 1.1 What is an Ordered Pair? . . . 4
- 1.2 Order Matters! . . . 4
- 1.3 More Examples of Ordered Pairs . . . 5
- 1.4 Equality of Ordered Pairs . . . 5

2 Cartesian Product: Definition and Computation 5
- 2.1 Definition . . . 5
- 2.2 Step-by-Step Method: The "Pairing" Approach . . . 6

3 Cardinality of Cartesian Product 7

4 Visual Representation of Cartesian Products 8
- 4.1 A) Table Representation . . . 8
- 4.2 B) Grid / Coordinate Representation . . . 8
- 4.3 C) Arrow Diagram . . . 9

5 Properties of Cartesian Product 9

6 In-Class Activities: Cartesian Product 9

PART B: Relations 11

7 Definition of a Relation 11
- 7.1 Why Do We Need Relations in Forensics? . . . 11

8 Examples of Relations 11

9 Domain, Range, and Codomain 13

10 Representation of Relations 13
- 10.1 A) Roster Form . . . 13
- 10.2 B) Set-Builder Form . . . 14
- 10.3 C) Arrow Diagram . . . 14
- 10.4 D) Table/Matrix Form . . . 14
- 10.5 E) Graph Form . . . 14

11 Types of Relations 15
- 11.1 Reflexive Relation . . . 15
- 11.2 Symmetric Relation . . . 15
- 11.3 Transitive Relation . . . 15
- 11.4 Equivalence Relation . . . 16

12 In-Class Activities: Relations 16

PART C: Functions **18**

13 Definition of a Function **18**
13.1 Terminology 18

14 Examples of Functions **18**

15 Types of Functions **19**
15.1 One-One (Injective) Function 19
15.2 Onto (Surjective) Function 20
15.3 Bijective Function 20

16 In-Class Activities: Functions **20**

17 Summary **21**

18 Practice Problems **22**

PART A: Cartesian Product of Sets

1 Ordered Pairs

1.1 What is an Ordered Pair?

Before we define the Cartesian product, we must first understand **ordered pairs**.

Definition (Ordered Pair): An **ordered pair** is a pair of elements written in a specific order:

$$(a, b)$$

where a is the **first element** (first coordinate) and b is the **second element** (second coordinate).

1.2 Order Matters!

$$(a, b) \neq (b, a) \quad \text{unless } a = b$$

This is **different** from sets, where order does not matter:

$$\{a, b\} = \{b, a\} \quad \text{(sets: order doesn't matter)}$$

$$(a, b) \neq (b, a) \quad \text{(ordered pairs: order matters!)}$$

Forensic Example: Suspect–Test Pair

A forensic lab has:

$$S = \{\text{Rahul},\ \text{Priya}\} \quad \text{(suspects)}$$

$$T = \{\text{DNA},\ \text{Fingerprint}\} \quad \text{(tests)}$$

The ordered pair $(\text{Rahul}, \text{DNA})$ means:

"Rahul was tested for DNA."

But $(\text{DNA}, \text{Rahul})$ would mean:

"DNA was tested for Rahul" — which makes no sense!

The **order** carries meaning: **first = person, second = test**.

1.3 More Examples of Ordered Pairs

Ordered Pair	Meaning
$(2, 5)$	$x = 2$, $y = 5$ on a graph
$(\text{Blood}, 2.1)$	Blood sample has toxin concentration 2.1 mg/L
$(\text{Scene A}, \text{knife})$	Knife was found at Scene A
$(\text{Suspect 3}, \text{match})$	Suspect 3's DNA matched
$(\text{Day 1}, 37.2°\text{C})$	Body temperature on Day 1 was 37.2°C

1.4 Equality of Ordered Pairs

Two ordered pairs are equal **if and only if** both corresponding elements are equal:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d$$

Example 1: Find x and y if $(2x + 1, 3y - 2) = (7, 10)$.

Solution:

$$2x + 1 = 7 \implies 2x = 6 \implies x = 3$$

$$3y - 2 = 10 \implies 3y = 12 \implies y = 4$$

So $x = 3$, $y = 4$.

Example 2: Find a and b if $(a + b,\ a - b) = (9, 3)$.

Solution:

$$a + b = 9 \quad \text{...(i)}$$

$$a - b = 3 \quad \text{...(ii)}$$

Adding (i) and (ii): $2a = 12 \implies a = 6$.
Subtracting (ii) from (i): $2b = 6 \implies b = 3$.
So $a = 6$, $b = 3$.

2 Cartesian Product: Definition and Computation

2.1 Definition

Definition (Cartesian Product): The **Cartesian product** of two sets A and B, written $A \times B$ (read as "A cross B"), is the set of **all possible ordered pairs** (a, b) where $a \in A$ and $b \in B$:

$$\boxed{A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}}$$

Why "Cartesian"? It is named after the French mathematician **René Descartes** (1596–1650), who introduced the coordinate system (the x-y plane) that we use today.

2.2 Step-by-Step Method: The "Pairing" Approach

Step 1. Take the **first element** of A.

Step 2. Pair it with **every element** of B.

Step 3. Move to the **next element** of A. Repeat.

Step 4. Collect all pairs into a set.

Example 3 (Basic Cartesian Product): Let $A = \{1, 2\}$ and $B = \{x, y, z\}$.
Step 1: Pair 1 with every element of B:

$$(1, x),\ (1, y),\ (1, z)$$

Step 2: Pair 2 with every element of B:

$$(2, x),\ (2, y),\ (2, z)$$

Step 3: Collect all pairs:

$$A \times B = \{(1, x),\ (1, y),\ (1, z),\ (2, x),\ (2, y),\ (2, z)\}$$

$$n(A \times B) = 6$$

Example 4: For the same sets, compute $B \times A$.
Solution:

$$B \times A = \{(x, 1),\ (x, 2),\ (y, 1),\ (y, 2),\ (z, 1),\ (z, 2)\}$$

Compare:

- $A \times B$ contains $(1, x)$
- $B \times A$ contains $(x, 1)$

These are **different** ordered pairs! So:

$$\boxed{A \times B \neq B \times A \quad \text{(in general)}}$$

Example 5 (Forensic Suspect–Test Matrix):
A forensic lab has:

$$S = \{\text{Arjun},\ \text{Neha}\} \quad \text{(suspects)}$$

$$T = \{\text{DNA},\ \text{Fingerprint},\ \text{Blood}\} \quad \text{(tests)}$$

Cartesian product $S \times T$:

$$S \times T = \begin{Bmatrix} (\text{Arjun}, \text{DNA}), & (\text{Arjun}, \text{Fingerprint}), & (\text{Arjun}, \text{Blood}), \\ (\text{Neha}, \text{DNA}), & (\text{Neha}, \text{Fingerprint}), & (\text{Neha}, \text{Blood}) \end{Bmatrix}$$

$$n(S \times T) = 2 \times 3 = 6$$

Interpretation: There are 6 possible **suspect–test combinations**. Each pair represents one test performed on one suspect.

Example 6 (Crime Scene $\times$ Evidence):

$$\text{Scenes} = \{\text{A}, \text{B}\}$$

$$\text{Evidence} = \{\text{knife},\ \text{gun},\ \text{rope}\}$$

$$\text{Scenes} \times \text{Evidence} = \begin{Bmatrix} (\text{A}, \text{knife}), & (\text{A}, \text{gun}), & (\text{A}, \text{rope}), \\ (\text{B}, \text{knife}), & (\text{B}, \text{gun}), & (\text{B}, \text{rope}) \end{Bmatrix}$$

Interpretation: This represents all possible "evidence found at a scene" combinations.

Example 7 (Cartesian Product with Empty Set): If $A = \{1, 2\}$ and $B = \varnothing$, then:

$$A \times B = \varnothing$$

because there are no elements in B to pair with. Similarly:

$$\boxed{A \times \varnothing = \varnothing \times A = \varnothing}$$

Example 8 (Set with Itself): Let $A = \{1, 2, 3\}$. Then $A \times A$ is:

$$A \times A = \begin{Bmatrix} (1,1), & (1,2), & (1,3), \\ (2,1), & (2,2), & (2,3), \\ (3,1), & (3,2), & (3,3) \end{Bmatrix}$$

$$n(A \times A) = 3 \times 3 = 9$$

Forensic Interpretation of $A \times A$: If A represents 3 crime scenes, then $A \times A$ represents all possible **scene-to-scene linkages** (including a scene linked to itself). This is used in **linkage analysis** to determine which scenes are connected.

3 Cardinality of Cartesian Product

Formula:

$$\boxed{n(A \times B) = n(A) \times n(B)}$$

If A has m elements and B has n elements, then $A \times B$ has mn elements.

Example 9:

$$n(A) = 5, \quad n(B) = 4 \implies n(A \times B) = 5 \times 4 = 20$$

Extension to Three Sets:

$$n(A \times B \times C) = n(A) \times n(B) \times n(C)$$

Example 10: If $n(A) = 2$, $n(B) = 3$, $n(C) = 4$, then:

$$n(A \times B \times C) = 2 \times 3 \times 4 = 24$$

Forensic Example: 2 suspects $\times$ 3 tests $\times$ 4 body fluids $=$ 24 possible test–sample combinations.

4 Visual Representation of Cartesian Products

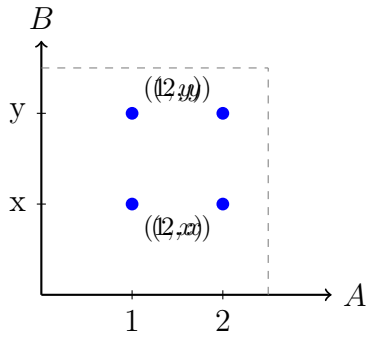
4.1 A) Table Representation

For $A = \{1, 2\}$ and $B = \{x, y, z\}$:

	x	y	z
1	$(1, x)$	$(1, y)$	$(1, z)$
2	$(2, x)$	$(2, y)$	$(2, z)$

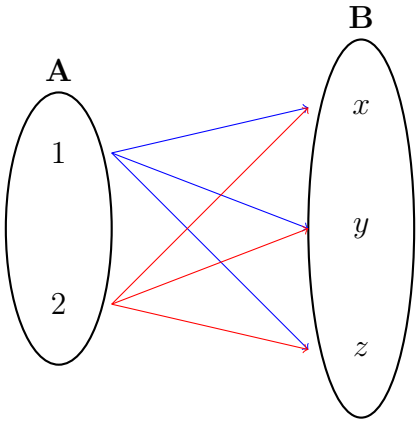
Each cell contains one ordered pair. This is also called a **matrix representation** of the Cartesian product.

4.2 B) Grid / Coordinate Representation



Each dot represents one ordered pair. This is exactly how we plot points on a graph!

4.3 C) Arrow Diagram



This shows all 6 pairings clearly. Each arrow from A to B represents one ordered pair in $A \times B$.

5 Properties of Cartesian Product

Property	Statement
Not Commutative	$A \times B \neq B \times A$ (in general)
Empty Set	$A \times \varnothing = \varnothing$
Cardinality	$n(A \times B) = n(A) \times n(B)$

6 In-Class Activities: Cartesian Product

Activity 1: Build the Cartesian Product (5 minutes, individual)

Let:

$$A = \{\text{Arjun},\ \text{Neha}\}, \quad B = \{\text{Blood},\ \text{DNA}\}$$

(a) Write out $A \times B$ in roster form.

(b) Find $n(A \times B)$.

(c) Write out $B \times A$.

(d) Is $A \times B = B \times A$? Explain.

Activity 2: Suspect–Weapon Matrix (10 minutes, pairs)

A forensic investigator has:

$$\text{Suspects} = \{\text{Ravi}, \text{Meena}, \text{Suresh}\}$$

$$\text{Weapons} = \{\text{knife}, \text{gun}\}$$

(a) Construct the Cartesian product Suspects $\times$ Weapons.

(b) How many possible suspect–weapon combinations are there?

(c) If the lab has 4 more suspects, how many combinations would there be?

(d) Draw an arrow diagram for this Cartesian product.

Activity 3: Cartesian Product with Three Sets (10 minutes, groups of 3)

Let:

$$A = \{\text{Scene A},\ \text{Scene B}\}$$

$$B = \{\text{knife},\ \text{gun}\}$$

$$C = \{\text{DNA},\ \text{Fingerprint},\ \text{Blood}\}$$

(a) How many elements are in $A \times B \times C$?

(b) Write out the first 4 ordered triples of $A \times B \times C$.

(c) If one suspect is linked to each triple, how many suspect–scene–weapon–test combinations exist?

Activity 4: True or False? (5 minutes, group discussion)

For any sets A, B, determine whether each statement is **always true**, **sometimes true**, or **never true**:

(1) $A \times B = B \times A$

(2) $n(A \times B) = n(A) \times n(B)$

(3) $A \times \varnothing = A$

(4) $(a, b) = (b, a)$ for all a, b

(5) $A \times A$ has $n(A)^2$ elements

(6) $\{1, 2\} \times \{3, 4\}$ contains $(1, 3)$ and $(3, 1)$

PART B: Relations

7 Definition of a Relation

Definition (Relation): A **relation** R from set A to set B is any **subset** of the Cartesian product $A \times B$.

$$\boxed{R \subseteq A \times B}$$

If $(a, b) \in R$, we write $a \; R \; b$ (read: "a is related to b").
If $(a, b) \notin R$, we write $a \not R \; b$.

Key Idea: The Cartesian product $A \times B$ contains **all possible** ordered pairs. A relation is any **selection** of some of those pairs, based on a rule or condition.

7.1 Why Do We Need Relations in Forensics?

In forensic science, we constantly deal with relationships:

- Which suspect matches which DNA sample?
- Which evidence belongs to which crime scene?
- Which drug was found in which body fluid?
- Which fingerprint matches which suspect?

Each of these is a **relation** between two sets.

8 Examples of Relations

Example 12 (Basic Relation): Let $A = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$.
Define a relation R as: "a is related to b if $b = 2a$".

(a) $1 \; R \; 2$ (since $2 = 2 \times 1$) ✓

(b) $2 \; R \; 4$ (since $4 = 2 \times 2$) ✓

(c) $3 \; R \; 6$ (since $6 = 2 \times 3$) ✓

So:

$$R = \{(1, 2),\ (2, 4),\ (3, 6)\}$$

Note: $R \subseteq A \times B$, and $n(R) = 3$, while $n(A \times B) = 9$.

Example 13 (Forensic Relation: Suspect–DNA Match):
Let:

$$S = \{\text{Suspect 1},\ \text{Suspect 2},\ \text{Suspect 3}\}$$

$$D = \{\text{DNA Sample A},\ \text{DNA Sample B}\}$$

Suppose:

- Suspect 1 matches DNA Sample A
- Suspect 3 matches DNA Sample B
- Suspect 2 does not match either sample

The relation R ("matches") is:

$$R = \{(\text{Suspect } 1, \text{DNA A}),\ (\text{Suspect } 3, \text{DNA B})\}$$

Interpretation: Only Suspects 1 and 3 are linked to DNA evidence. Suspect 2 is excluded based on DNA comparison.

Example 14 (Forensic Relation: Evidence–Scene):

Let:

$$E = \{\text{knife},\ \text{gun},\ \text{rope}\}$$

$$\text{Scenes} = \{\text{A},\ \text{B}\}$$

Define R: "evidence e was found at scene s".

Suppose:

- Knife was found at Scene A
- Gun was found at both Scene A and Scene B
- Rope was found only at Scene B

$$R = \{(\text{knife}, \text{A}),\ (\text{gun}, \text{A}),\ (\text{gun}, \text{B}),\ (\text{rope}, \text{B})\}$$

Interpretation: The gun appears at both scenes, which could link the two scenes together. The knife and rope are unique to their respective scenes.

Example 15 (Forensic Relation: Drug–Sample):

Let:

$$\text{Drugs} = \{\text{Cocaine},\ \text{Heroin},\ \text{Cannabis}\}$$

$$\text{Samples} = \{\text{Blood},\ \text{Urine}\}$$

Define R: "drug d was detected in sample s".

Suppose:

- Cocaine was found in Blood (not Urine)
- Heroin was found in Urine (not Blood)
- Cannabis was found in both

$$R = \{(\text{Cocaine}, \text{Blood}),\ (\text{Heroin}, \text{Urine}),\ (\text{Cannabis}, \text{Blood}),\ (\text{Cannabis}, \text{Urine})\}$$

Interpretation: Cannabis is the most widely detected drug across both sample types. Heroin is only detectable in urine (suggesting recent use).

9 Domain, Range, and Codomain

For a relation $R \subseteq A \times B$:

- **Domain:** The set of all **first elements** of the ordered pairs in R.

$$\text{Dom}(R) = \{a \in A \mid (a, b) \in R \text{ for some } b \in B\}$$

- **Range:** The set of all **second elements** of the ordered pairs in R.

$$\text{Range}(R) = \{b \in B \mid (a, b) \in R \text{ for some } a \in A\}$$

- **Codomain:** The entire set B (all possible outputs).

Example 16: For $R = \{(1, 2),\ (2, 4),\ (3, 6)\}$ from $A = \{1, 2, 3\}$ to $B = \{2, 4, 6, 8\}$:

- $\text{Dom}(R) = \{1, 2, 3\}$
- $\text{Range}(R) = \{2, 4, 6\}$
- Codomain $= B = \{2, 4, 6, 8\}$

Notice: Range $\subseteq$ Codomain (6 is in the range, 8 is not).

Example 17 (Forensic Interpretation): Using Example 13:

$$R = \{(\text{Suspect } 1, \text{DNA A}),\ (\text{Suspect } 3, \text{DNA B})\}$$

- $\text{Dom}(R) = \{\text{Suspect } 1,\ \text{Suspect } 3\}$
- $\text{Range}(R) = \{\text{DNA A},\ \text{DNA B}\}$

Interpretation: The domain tells us **which suspects matched**, and the range tells us **which DNA samples were matched**. Suspect 2 is not in the domain (no match), so they are excluded.

10 Representation of Relations

Relations can be represented in several ways:

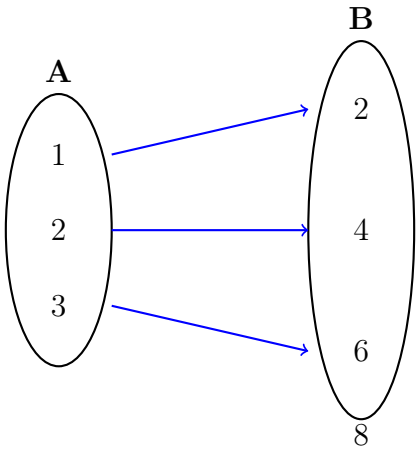
10.1 A) Roster Form

$$R = \{(1, 2),\ (2, 4),\ (3, 6)\}$$

10.2 B) Set-Builder Form

$$R = \{(a, b) \in A \times B \mid b = 2a\}$$

10.3 C) Arrow Diagram

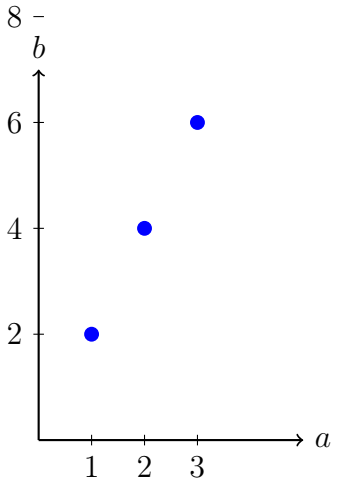


10.4 D) Table/Matrix Form

	2	4	6	8
1	✓			
2		✓		
3			✓	

A checkmark at position (i, j) means $(i, j) \in R$.

10.5 E) Graph Form



This is a **scatter plot** representation of the relation.

11 Types of Relations

11.1 Reflexive Relation

A relation R on a set A is **reflexive** if **every** element of A is related to itself:

$$(a, a) \in R \quad \text{for all } a \in A$$

Example 18: Let $A = \{1, 2, 3\}$.
$R_1 = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$ is **reflexive** because all diagonal pairs $(1, 1), (2, 2), (3, 3)$ are present.
$R_2 = \{(1, 1), (2, 2)\}$ is **not reflexive** because $(3, 3)$ is missing.

Forensic Application: In a **similarity matrix** of evidence samples, every sample is trivially similar to itself (self-similarity = 1). So a similarity relation is always reflexive.

11.2 Symmetric Relation

A relation R on a set A is **symmetric** if:

$$(a, b) \in R \implies (b, a) \in R$$

Example 19: Let $A = \{1, 2, 3\}$.
$R_1 = \{(1, 2), (2, 1), (2, 3), (3, 2)\}$ is **symmetric**.
$R_2 = \{(1, 2), (2, 3)\}$ is **not symmetric** because $(1, 2) \in R$ but $(2, 1) \notin R$.

Forensic Application: If Sample A is **similar to** Sample B, then Sample B is similar to Sample A. So similarity relations are **symmetric**. Distance matrices used in forensic comparison are always symmetric.

11.3 Transitive Relation

A relation R on a set A is **transitive** if:

$$(a, b) \in R \text{ and } (b, c) \in R \implies (a, c) \in R$$

Example 20: Let $A = \{1, 2, 3\}$.
$R_1 = \{(1, 2), (2, 3), (1, 3)\}$ is **transitive**.
$R_2 = \{(1, 2), (2, 3)\}$ is **not transitive** because $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R$.

Forensic Application: If Suspect A's DNA matches Scene 1, and Scene 1's evidence matches Scene 2, then Suspect A is linked to Scene 2. This is **transitive linkage**.

11.4 Equivalence Relation

A relation R is an **equivalence relation** if it is:

(i) **Reflexive**

(ii) **Symmetric**

(iii) **Transitive**

Example 21: On $A = \{1, 2, 3\}$, the relation:

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$$

is an equivalence relation. It partitions A into $\{1, 2\}$ and $\{3\}$.

Forensic Application: Blood group compatibility is not an equivalence relation (it is not symmetric). But **"same blood type"** is an equivalence relation — it partitions people into groups (A, B, AB, O).

12 In-Class Activities: Relations

Activity 5: Suspect–Evidence Relation (10 minutes, pairs)

Let:

$$\text{Suspects} = \{\text{S1}, \text{S2}, \text{S3}\}$$

$$\text{Evidence} = \{\text{Fingerprint}, \text{DNA}, \text{Fiber}\}$$

Suppose:

- S1 matches Fingerprint and DNA
- S2 matches Fiber only
- S3 matches nothing

(a) Write the relation R in roster form.

(b) Find Dom(R) and Range(R).

(c) Is R reflexive? Symmetric? Transitive? Explain.

(d) Draw an arrow diagram.

Activity 6: Reflexive, Symmetric, Transitive (10 minutes, groups of 3)

For each relation below on $A = \{1, 2, 3\}$, determine whether it is reflexive, symmetric, and/or transitive.

(a) $R_1 = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$

(b) $R_2 = \{(1,2),(2,3),(1,3)\}$

(c) $R_3 = \{(1,1),(1,2),(2,1),(2,2),(3,3)\}$

(d) $R_4 = \{(1,1),(2,2),(3,3)\}$

(e) $R_5 = \{(1,2),(2,1),(1,3),(3,1)\}$

PART C: Functions

13 Definition of a Function

Definition (Function): A **function** f from set A to set B, written $f : A \to B$, is a **special type of relation** in which:

(i) **Every** element of A is paired with **at least one** element of B.

(ii) **No** element of A is paired with **more than one** element of B.

In other words: **each input has exactly one output.**

Function vs Relation: Every function is a relation, but not every relation is a function. A function is a relation where each input has exactly one output.

13.1 Terminology

- **Domain:** The set A (all possible inputs).
- **Codomain:** The set B (all possible outputs).
- **Range:** The set of **actual** outputs (subset of codomain).
- **Image:** If $f(a) = b$, then b is the image of a under f.
- **Pre-image:** a is the pre-image of b.

Forensic Example: Height Estimation from Femur Length

Let:

$$A = \{\text{set of femur lengths (cm)}\}$$

$$B = \{\text{set of possible heights (cm)}\}$$

The anthropologist's formula $f(x) = 2.6x + 50$ is a function:

- Each femur length x gives **exactly one** estimated height.
- A 40 cm femur $\to$ height $= 2.6(40) + 50 = 154$ cm.
- A 45 cm femur $\to$ height $= 2.6(45) + 50 = 167$ cm.

This is a valid function because each input produces a unique output.

14 Examples of Functions

Example 22 (Function): Let $A = \{1, 2, 3\}$ and $B = \{2, 4, 6\}$.

Define $f : A \to B$ by $f(x) = 2x$. Then:

$$f(1) = 2, \quad f(2) = 4, \quad f(3) = 6$$

$$f = \{(1,2),(2,4),(3,6)\}$$

This is a function because:

- Every element in A has an output ✓
- No element in A has two outputs ✓

Example 23 (Not a Function): Let $A = \{1,2,3\}$, $B = \{2,4,6\}$.

$$R = \{(1,2),(1,4),(2,6),(3,2)\}$$

This is **NOT** a function because:

- The input 1 is paired with **two** outputs (2 and 4). ×

Example 24 (Not a Function): Let $A = \{1,2,3\}$, $B = \{2,4,6\}$.

$$R = \{(1,2),(2,4)\}$$

This is **NOT** a function from A to B because:

- The element 3 in A has no output. ×

Example 25 (Forensic Function): DNA Profile Match Probability

Let:

$$A = \{\text{set of all DNA profiles}\}$$

$$B = \{\text{set of match probabilities (0 to 1)}\}$$

Define $f : A \to B$ by $f(\text{profile}) = \text{probability of random match.}$

Each DNA profile has exactly one match probability in a given population. This is a valid function.

15 Types of Functions

15.1 One-One (Injective) Function

A function $f : A \to B$ is **one-one (injective)** if **different inputs** give **different outputs**:

$$f(a_1) = f(a_2) \implies a_1 = a_2$$

Equivalently: no two different elements of A map to the same element of B.

Example 26: $f : \{1,2,3\} \to \{2,4,6\}$ defined by $f(x) = 2x$ is **one-one**.

Because $f(1) = 2$, $f(2) = 4$, $f(3) = 6$ — all outputs are different. ✓

Example 27 (Not One-One): $f : \{1, 2, 3\} \to \{2, 4\}$ defined by:

$$f(1) = 2, \quad f(2) = 4, \quad f(3) = 2$$

This is **not one-one** because $f(1) = f(3) = 2$. $\times$

Forensic Application: A **one-one** relationship between fingerprints and individuals means each fingerprint belongs to exactly one person and no two people share the same fingerprint. This is the basis of **fingerprint identification**.

15.2 Onto (Surjective) Function

A function $f : A \to B$ is **onto (surjective)** if **every** element of B is the image of at least one element of A:

$$\text{Range}(f) = B$$

Example 28: $f : \{1, 2, 3\} \to \{2, 4, 6\}$ defined by $f(x) = 2x$ is **onto** because every element of B (2, 4, 6) is hit. ✓

Example 29 (Not Onto): $f : \{1, 2, 3\} \to \{2, 4, 6, 8\}$ defined by $f(x) = 2x$.

Here, $\text{Range}(f) = \{2, 4, 6\}$, but Codomain $= \{2, 4, 6, 8\}$. Since 8 is not hit, f is **not onto**. $\times$

15.3 Bijective Function

A function is **bijective** if it is **both one-one and onto**. Bijective functions establish a **perfect pairing** between the two sets.

Forensic Application: A bijection between suspects and unique DNA profiles would mean that each suspect has a unique profile AND every profile belongs to a suspect. In practice, real DNA databases use **probabilistic matching**, but the mathematical ideal is a bijection.

16 In-Class Activities: Functions

Activity 7: Function or Not? (5 minutes, individual)

For each relation below from $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$, determine whether it is a function. Explain.

(a) $R_1 = \{(1, a), (2, b), (3, c)\}$

(b) $R_2 = \{(1, a), (1, b), (2, c), (3, a)\}$

(c) $R_3 = \{(1, a), (2, a), (3, a)\}$

(d) $R_4 = \{(1, a), (2, b)\}$

(e) $R_5 = \{(1, a), (2, b), (3, b)\}$

Activity 8: One-One, Onto, Bijective (10 minutes, pairs)

For each function below, determine whether it is one-one, onto, or bijective. Justify.

(a) $f : \{1, 2, 3\} \to \{2, 4, 6\}$, $f(x) = 2x$

(b) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$

(c) $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 1$

(d) $f : \{1, 2, 3, 4\} \to \{a, b, c, d\}$, $f = \{(1, a), (2, b), (3, c), (4, d)\}$

(e) $f : \{1, 2, 3\} \to \{a, b\}$, $f = \{(1, a), (2, b), (3, a)\}$

Activity 9: Height Estimation (10 minutes, individual)

The height estimation formula from femur length is:

$$f(x) = 2.6x + 50$$

where x = femur length (cm), $f(x)$ = estimated height (cm).

(a) If a femur is 42 cm, what is the estimated height?

(b) If a femur is 38 cm, what is the estimated height?

(c) Is this function one-one? Explain.

(d) Sketch the graph of $f(x)$ for x from 30 to 50.

(e) If remains of a person are found and the estimated height is 175 cm, what was the femur length?

Activity 10: Time of Death Estimation (10 minutes, groups of 3)

A body is found at 6:00 AM with a temperature of 28°C. The room temperature is 20°C. The cooling constant is $k = 0.15$ per hour. Assume $T_0 = 37$°C.

Using Newton's Law of Cooling:

$$T(t) = 20 + (37 - 20)e^{-0.15t}$$

(a) Solve for t when $T(t) = 28$.

(b) Interpret your answer: how long has the person been dead?

(c) At what time did the person likely die?

(d) Is this a one-one function? Why does that matter?

17 Summary

Key Takeaways

1. **Ordered pair** (a, b): order matters.
2. **Cartesian product** $A \times B$: set of all ordered pairs (a, b).
3. $n(A \times B) = n(A) \times n(B)$.
4. **Relation** $R \subseteq A \times B$: any subset of the Cartesian product.
5. **Domain:** first elements; **Range:** second elements; **Codomain:** entire set B.
6. **Types of relations:** reflexive, symmetric, transitive, equivalence.
7. **Function:** a relation where each input has exactly one output.
8. **Types of functions:** one-one (injective), onto (surjective), bijective.
9. **Forensic applications:** suspect–test combinations, crime scene linkage, evidence classification, height estimation, time of death.

18 Practice Problems

1. If $A = \{1, 2, 3\}$ and $B = \{a, b\}$, find:
   (a) $A \times B$
   (b) $B \times A$
   (c) $n(A \times B)$
   (d) Is $A \times B = B \times A$?
2. Find x and y if $(x + 2, y - 3) = (5, 7)$.
3. Let $A = \{1, 2, 3, 4\}$. Define $R = \{(a, b) \mid a + b = 6\}$.
   (a) Write R in roster form.
   (b) Find Dom(R) and Range(R).
   (c) Is R reflexive? Symmetric? Transitive?
4. For each relation on $A = \{1, 2, 3\}$, determine whether it is a function:
   (a) $R_1 = \{(1, 1), (2, 2), (3, 3)\}$
   (b) $R_2 = \{(1, 2), (1, 3), (2, 2), (3, 1)\}$
   (c) $R_3 = \{(1, 1), (2, 1), (3, 1)\}$
   (d) $R_4 = \{(1, 2), (2, 3)\}$
5. Determine whether $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 5x - 3$ is:
   (a) One-one
   (b) Onto
   (c) Bijective

6. **(Forensic Application)** A forensic anthropologist uses the formula $H = 2.5F + 55$, where F is femur length (cm) and H is height (cm).
   (a) Find the estimated height for $F = 40$, $F = 44$, $F = 48$.
   (b) Is this a function? Why?
   (c) Is it one-one? Justify your answer.
   (d) If a skeleton has an estimated height of 165 cm, find the femur length.

7. **(Forensic Application)** The time since death (in hours) is given by:

$$t = -\frac{1}{0.12} \ln\left(\frac{T - 22}{15}\right)$$

   where T is the current body temperature (°C).
   (a) Find t when $T = 30$°C.
   (b) Find t when $T = 25$°C.
   (c) Is t a function of T? Explain.

— End of Notes —